

Question One: Consider the following schema for a company (assume attributes do not take a null value)

```
employee( eid, name, age, did)
department(did, dept_name, location)
```

- According to the given schema and assumption, can an employee work in more than one department?
- Please draw an ER-diagram to describe the relationship between employee and department.

Question Two: *Consider the following the following business data application, your team is tasked with database application development. You're in charge of the first important step: conceptual data modelling. Please produce an E/R diagram for your team. You've recorded the following data requirements. Human language is messy, but you'll faithfully, and accurately express the semantics.*

Note: Please use the main drawing conventions in the lecture notes.

1. A course is uniquely identified by its course ID. Data about each course are: course name, lecturer name, maximum capacity, and total number of students who enrol in each course as well.
2. Each course can have zero or more labs.
3. A lab is uniquely identified by a lab ID. Each lab is described by these data: time, weeks, and one location (composed of one building-name and one room-number).
4. A lab can be scheduled for multiple weeks.
5. A student is uniquely identified by his/her Sid.
6. Each student has attributes: name, contact number, email and degree.
7. A lab tutor is uniquely identified by his/her ID.
8. Each lab tutor has attributes: name, phone no., email and salary.
9. A student can enrol in zero or more courses, and can enrol in zero or more labs.
10. Each lab must belong to exactly one course, and must have at least one tutor on duty.
11. A lab tutor can work for zero or more courses, and zero or more labs.
12. A course or a lab must have at least one student.
13. A course can have multiple lab tutors but is not compulsory to have lab tutors.

Question Two (Extra): Translate the ER diagram of the above question into a relational schema.

Question Three: Please translate an ER diagram into a relational diagram

